Quality Control Layer 1: Research & Factual Accuracy Audit

Lesson: HTTP Requests and Responses | **Audit Reference:** AUDIT-FACT-001

Overall Audit Status: NEEDS IMPROVEMENT **Factual Accuracy Score:** 94.0% **Issues Detected:** 4

1. Executive Accuracy Assessment

The lesson accurately reflects RFC 9110 and MDN Web Docs regarding HTTP methods (GET/POST), idempotency, safety, caching, and 2xx/4xx/5xx status code categories. Specific status code definitions (200, 201, 204, 400, 401, 403, 404, 429, 500, 502, 503, 504) are consistent with established standards. However, research integrity is weakened by citing a preview-only book source for factual claims, and one technical simplification (403 Forbidden) could cause minor misconceptions about authentication versus authorization.

2. Factual Claims Analysis & Corrections (4 items)

ID / Severity	Affected Section & Claim	Audit Finding & Supporting Evidence	Recommended Correction
FACT-001 MEDIUM	**Section:** Sections 3, 4, 10, 11 (General Citations & References) *"Sources: [RFC 9110: HTTP Semantics], [MDN Web Docs], [HTTP: The Definitive Guide] and subsequent citations to [HTTP: The Definitive Guide] throughout the lesson."*	Source [3] (HTTP: The Definitive Guide) is accessible only as Metadata/Preview per the provided research sources. The lesson cites it as a reference for factual claims without indicating that full-text verification was unavailable. This presents a source reliability and attribution issue because claims attributed to this source cannot be independently verified from the provided materials. **Evidence:** Source [3] access level: Metadata/Preview only (Google Books). RFC 9110 and MDN are Full Text.	**Fix:** Remove [HTTP: The Definitive Guide] from factual citations unless full-text access is obtained, or add a disclaimer that this source was not fully verifiable. Prioritize RFC 9110 and MDN for all technical claims.

FACT-002 LOW	**Section:** Section 4 - Status Code Categories (4xx Client Error) *"403 Forbidden: The client is authenticated but does not have permission."*	RFC 9110 Section 15.5.3 defines 403 Forbidden as the server understanding the request but refusing to authorize it. It does not require that the client be authenticated; 401 Unauthorized is the code for missing or failed authentication. Stating that the client is authenticated conflates authentication with authorization and may mislead learners about when to use 401 versus 403. **Evidence:** RFC 9110 Section 15.5.3; MDN Web Docs: HTTP response status codes.	**Fix:** 403 Forbidden: The server understood the request but refuses to authorize it. (Use 401 for missing or failed authentication.)
FACT-003 LOW	**Section:** Available Research Sources & Section 11 References *"Source [4]: HTTP Crash Course & Status Codes Explained ... Accessible: Full Text"*	A YouTube video is not a full-text document. While a transcript may be available, classifying it as 'Full Text' equivalent to RFC 9110 or MDN overstates its textual verifiability and research utility. **Evidence:** Source [4] is a YouTube video; full-text accessibility is limited to transcript/captions if provided.	**Fix:** Reclassify Source [4] accessibility as 'Video / Transcript (if available)' rather than 'Full Text'.
FACT-004 LOW	**Section:** Section 4 - GET vs POST Table *"Body allowed: No (typically) [for GET]"*	RFC 9110 Section 9.3.1 states that a payload within a GET request has no defined semantics and that some existing implementations might reject it, but the specification does not strictly prohibit a body. 'No (typically)' is directionally correct but imprecise for a technical audit. **Evidence:** RFC 9110 Section 9.3.1.	**Fix:** Body allowed: Not defined by RFC 9110 (typically ignored or rejected by implementations)

3. Verification Scope & Limitations

Source [3] (HTTP: The Definitive Guide) is available only as a Google Books preview/metadata; full-page text verification was not possible. Source [4] is a YouTube video; verification is limited to publicly available transcript/captions rather than a structured document. All RFC 9110 and MDN claims were verified against full-text accessible sources.

4. Conclusion & Certification

Overall, the lesson is factually accurate for its stated learning objectives. Recommended changes are minor and relate to source attribution precision and one definitional refinement for 403 Forbidden.